



Voxplorer: Voice data exploration and projection in an interactive dashboard

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Abstract

The acoustic signal of voice is inherently multi-dimensional. Early speech and voice research often focused on isolated or limited acoustic features. However, past research has demonstrated that a comprehensive understanding of voice requires analysing multiple dimensions simultaneously. The Voxplorer interactive dashboard addresses this issue by making state-of-the-art feature extraction and dimensionality reduction methods more accessible. It allows users to interactively explore, subset, and visualise pre-computed high-dimensional data, or extract features from recordings directly in the dashboard. The Voxplorer aims to provide modern technical resources for researchers, broadening the ideas and scope of future research in the field of voice communication sciences.

Index Terms: interactive dashboard, explorative data analysis, feature extraction

1. Introduction

The acoustic signal of voice is inherently multi-dimensional. Early speech and voice research often focused on isolated or limited acoustic features—such as fundamental frequency (f0) in intonation studies, syllable duration in rhythm research, or formants (F1 and F2) in vowel quality analysis. However, past research has demonstrated that a comprehensive understanding of voice requires analysing multiple dimensions simultaneously. Tools like Voice Sauce [1] extract a broad range of psychoacoustically relevant vocal parameters, while other approaches rely on abstract acoustic representations, such as Mel-Frequency Cepstral Coefficients (MFCCs). Additionally, advanced speaker characterization methods, including i-vectors [2] and x-vectors [3], transform a speaker's voice into hundreds of abstract dimensions specifically optimized for speaker identity. These high-dimensional embeddings not only capture speaker-specific traits but also encode valuable para-linguistic information, such as emotions and speaking style [4, 5]. Extracting and interpreting such multi-dimensional voice data presents significant challenges. Here, we introduce Voxplorer, a tool designed to facilitate the exploration and analysis of multi-dimensional vocal characteristics. One of our inspirations for our dashboard is Tensorflow's TensorBoard embedding projector. The embedding projector allows users to visualise data embeddings in a reduced dimensions space in an interactive figure. Unfortunately, it does not allow users to download subsets of the data created through filtering or interactive selection within the plot.

With the Voxplorer dashboard we aim to provide an all-in-one tool, providing feature extraction, dataset interactive exploration, dimensionality reduction using different algorithms, and

interactive visualisation. Our goal is to empower speech scientists with access to modern methods and techniques to analyse complex feature sets and corpora.

2. Workflow

The workflow is described using a block diagram in Figure 1. The main workflow of the dashboard is the "visualiser". Users are first presented with two options:

1. Upload table: upload a table of pre-computed features (CSV, TSV, or Excel file).
2. Upload audio: upload a list of audio files (WAV files).

For the first option, users select columns containing metavariables (labels for observations) after uploading the table. For the second option, users specify the feature extraction method and parameters for the audio files (more details in section 2.1). They can also define metavariables from filenames (e.g. `speakerID.sex.speakingstyle.wav` can be split by "-" to the variables `speakerID`, `sex`, and `speakingstyle`).

The dashboard will display the uploaded or computed table in an interactive table component, allowing the user to select rows and filter by values. The user will also be able to specify which dimensionality reduction algorithm to use and any options related to the chosen algorithm in a form. A description of the dimensionality reduction algorithms is provided in section 2.2. The dashboard will then compute the embeddings of the data features in the reduced space and display an interactive figure. Any metavariable that was specified during the data processing process will be available as an option to label by colour and/or by shape the data points. Any selection made from within the figure using the box- or lasso- select tools, or made from the table, is synchronised between the interactive table and the interactive figure.

Users are finally able to export the figure as an image file and both the original or computed, and reduced embeddings datasets as CSV files. The table exports can be made for either the full dataset as well as for only the currently selected data.

2.1. Feature extraction methods

For feature extraction methods we decided to focus on methods that are reliable when applied in an unsupervised manner to a vast variety of data. Although, this limits the amount of options we can provide, it does ensure that the feature extraction output quality is consistent for all users and corpora (apart from signal quality-related differences). We settled on pro-

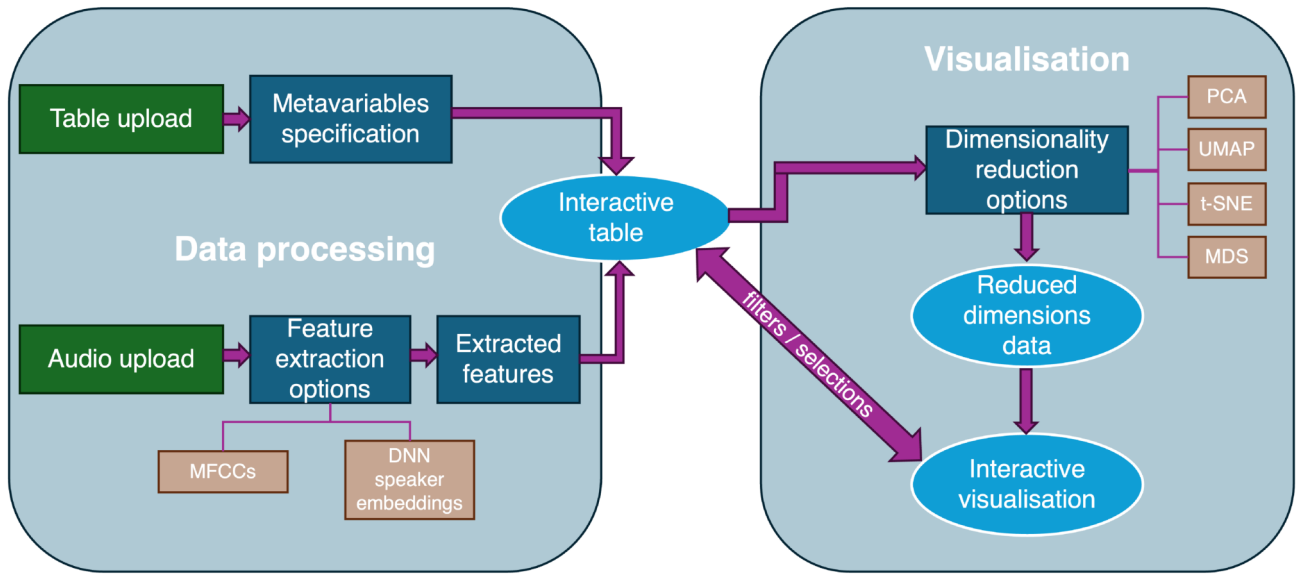


Figure 1: Block diagram of main dashboard workflow.

viding MFCCs (and both derivatives of the coefficients) supported by `librosa` [6]. Additionally, we provide support for DNN speaker embeddings extraction using pre-trained models through `SpeechBrain` [7].

2.2. Dimensionality reduction methods

We decided to give users access to common linear and non-linear dimensionality reduction methods. The dimensionality reduction methods we provide are:

- Principal Components Analysis (PCA) [8] through `scikit-learn` [9].
- Multi-dimensional Scaling (MDS) [10] through `scikit-learn` [9].
- t-distributed Stochastic Neighbour Embedding (t-SNE) [11] through `scikit-learn` [9].
- Uniform Manifold Approximation and Projection (UMAP) [12] through `umap-learn` [12].

2.3. Software and resource requirements

The dashboard is fully built in Python using `plotly-dash`. It idles at approximately between 3 and 5 GB in memory consumption when working with a 1800 observations dataset, but memory and computational resources consumption is dependent on choice of algorithms as well.

3. Outlook

With the Voxplorer dashboard we aim to empower voice researchers with access to modern analytical tools and a platform to interactively explore their large and complex speech corpora. The dashboard will be made available to use online and as stand-alone software at the following repository: <https://github.com/liri-uzh/voxplorer>.

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